

Homework (Habanero)

Q1. Solve for x in the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. What is the solution to $|2x - 3| = 7$?

- (A) $x = 2$ or $x = -5$
- (B) $x = 5$ or $x = -2$
- (C) $x = 1$ or $x = -4$
- (D) $x = 3$ or $x = -1$
- (E) $x = 4$ or $x = -3$

Q3. A patient's dosage of medicine is given by $|d - 10| \leq 2$, where d is the dosage in milligrams. What is the range of possible dosages?

- (A) $8 \leq d \leq 12$
- (B) $6 \leq d \leq 14$
- (C) $10 \leq d \leq 15$
- (D) $5 \leq d \leq 10$
- (E) $12 \leq d \leq 18$

Q4. The growth of a cell population is modeled by $|p - 20| = 5$, where p is the population size. What are the possible population sizes?

- (A) $p = 15$ or $p = 25$
- (B) $p = 10$ or $p = 30$
- (C) $p = 5$ or $p = 35$
- (D) $p = 0$ or $p = 40$
- (E) $p = 20$

Q5. Solve for x in the equation $|x + 2| = |3x - 1|$.

- (A) $x = 1$
- (B) $x = -1$
- (C) $x = 3$ or $x = -1$
- (D) $x = 2$ or $x = -2$
- (E) $x = 0$

Q6. A medical statistician analyzes data on the absolute difference in blood pressure between two groups of patients, given by $|b - 120| = 10$. What is the range of possible blood pressures?

- (A) $110 \leq b \leq 130$
- (B) $100 \leq b \leq 140$
- (C) $90 \leq b \leq 150$
- (D) $80 \leq b \leq 160$
- (E) $70 \leq b \leq 170$

Q7. The concentration of a medication in a patient's bloodstream is given by $|c - 5| \leq 1$. What is the range of possible concentrations?

- (A) $4 \leq c \leq 6$
- (B) $3 \leq c \leq 7$
- (C) $2 \leq c \leq 8$
- (D) $1 \leq c \leq 9$
- (E) $0 \leq c \leq 10$

Q8. A biologist studies the growth of a microorganism, modeled by $|g - 10| = 3$. What are the possible growth rates?

- (A) $g = 7$ or $g = 13$
- (B) $g = 5$ or $g = 15$
- (C) $g = 3$ or $g = 17$
- (D) $g = 1$ or $g = 19$
- (E) $g = 0$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $|2x + 5| = |x - 3|$. Show all steps and check for extraneous solutions. Required work:

Q10. A pharmaceutical company is testing a new medication. The dosage is given by $|d - 15| \leq 5$. Find the range of possible dosages and explain the meaning of the absolute value in this context. Required work:

Chef's Tips

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- Tip 1: Emphasize the importance of checking for extraneous solutions.
- Tip 2: Encourage students to use graphs to visualize the solutions.
- Tip 3: Remind students to consider the context of the problem when interpreting the results.